PARTH PAREKH

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EDUCATION

Rutgers University – New Brunswick, NJ
B.S. in Computer Science | Minor in Data Science | May 2026
Relevant Coursework: Intro to Artificial Intelligence, Design and Analysis of Computer Algorithms, Data Structures, Software Methodology, Systems Programming, Computer Architecture, Data Management, Machine Learning Principles

EXPERIENCE

Data Analytics Intern – DOWC *June 2026 – Present*
Parsippany, NJ (Onsite)
- Built a reusable Apache Airflow ETL framework for SFTP-to-PostgreSQL pipelines, standardizing DAG architecture, configuration patterns, and shared utilities across production workflows serving major client data feeds
- Leveraged Snowflake's columnar architecture and micro-partitioning to optimize analytical workloads, reducing complex query times from 20+ minutes to under 1 minute on multi-million row datasets
- Built an end-to-end automation pipeline using Power Automate and Python to detect incoming vendor emails, extract and standardize CPO data across Excel file formats, and load structured records into the production database

Technical Business Analyst Intern – Marketeq Digital *Sept 2024 – Feb 2025*
Miami, FL (Remote)
- Bridged engineering and business stakeholders: translated wireframes into user stories and acceptance criteria, enabling modular feature rollouts across 3 product teams
- Mapped data integration flows across Strapi CMS, MongoDB, and Customer.io on Figma to support personalized automated email campaigns for 150+ clients

PROJECTS

GitHub Review Bot | github.com/pvparekh/github-review-bot *Apr 2026 – May 2026*
- Developed an autonomous GitHub App AI code reviewer using Python, FastAPI, and Claude that processes pull requests and posts structured inline comments within ~10 seconds of a push
- Secured webhook server with HMAC-SHA256 signature verification and RSA JWT authentication; built unified diff parser feeding a two-pass Claude pipeline that returns structured JSON per line and a markdown PR summary
- Containerized and deployed the application using Docker and AWS EC2; configured GitHub Actions CI/CD for automated redeployment workflows

Formula Vision | github.com/pvparekh/F1-Viewer | formulavision.vercel.app *Dec 2025 – May 2026*
- Built a full-stack F1 race replay platform with React, TypeScript, FastAPI, and WebSockets; engineered 60 FPS interpolated animation from 12.5 FPS server data using SVG viewBox transformations and requestAnimationFrame, streaming 440MB of historical telemetry across 5 Grand Prix races
- Optimized memory footprint from 600MB to 150MB for Render's 512MB free tier through LRU cache eviction with gc.collect(), strategic 30% frame truncation (95MB to 66MB files), and concurrent connection limiting, enabling efficient production deployment
- Integrated FastF1 API for verified race results; implemented dynamic camera follow system with 2.5x zoom tracking, playback controls (play/pause/seek, 4x speed, ±10s skip), real-time grid and driver telemetry panel

AetherFlow: Expense Intel | github.com/pvparekh/aetherflow | aetherflow-three.vercel.app *July 2025 – Apr 2026*
- Architected and shipped a full-stack SaaS-style expense intelligence platform: Next.js 15 App Router, TypeScript, Tailwind CSS, Supabase (PostgreSQL + Auth), deployed to Vercel; supports CSV, PDF, and TXT uploads
- Designed a two-pass AI pipeline: GPT-mini batch-categorizes 25 transactions per call into 9 categories; GPT-4o asynchronously generates a financial health score, insights, and savings opportunities from pre-aggregated statistics
- Built deterministic analytics engine with per-transaction z-scores, MAD z-scores, rolling category averages, drift detection (>15% shift), and vendor intelligence at zero inference cost

SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, R, HTML/CSS
Frontend: React, Next.js 15, Tailwind CSS, Framer Motion, Recharts
Backend: FastAPI, Node.js, REST APIs, WebSockets, Webhooks, HMAC-SHA256, JWT/OAuth, Async processing
AI / APIs: Claude, OpenAI GPT-4o, LLM API integration, Prompt engineering, Structured JSON output
Databases: PostgreSQL, Supabase, Snowflake,, MongoDB
Tools: Git, Docker, AWS (EC2), Apache Airflow, GitHub Actions (CI/CD), GitHub Apps API, Vercel, Render, Power BI, Unix/Linux, Microsoft Power Automate